Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details



a. Team name: VibeRick
b. Team leader name: Jaeho Shin
c. Problem Statement: Media, Content & Digital Experiences — creators reach only the viewers who speak their language. VibeRick Dub turns one video into dubbed tracks in the creator's own voice.

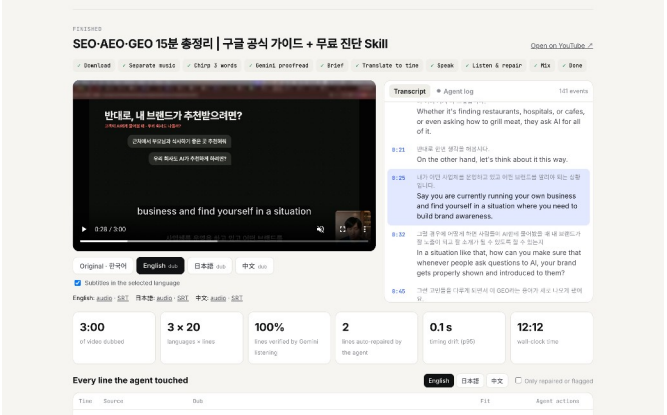
Brief about the idea

One video. Every language. Your own voice.

Paste a YouTube link. An AI agent built on Gemini transcribes it, translates every sentence to fit its timing, speaks it in the creator's own cloned voice, then listens back to every line and repairs what doesn't sound right.

The creator gets dubbed tracks and subtitles ready for YouTube's multi-language audio — in minutes, for cents per minute, with consent built in: the cloned voice is used only for the person who recorded it.

Languages today: Korean · English · Japanese · Chinese (source detected automatically).



Opportunities

How is it different from existing ideas?

- YouTube auto-dubbing: not the creator's voice, cannot be edited, and Korean sources get only English and Indonesian. VibeRick Dub speaks in the creator's own voice, in any of our languages, and shows every line.
- Voice-clone dubbing services (ElevenLabs, Rask, Perso) charge ~$0.33–2.20 per minute and run open-loop. Our agent listens to every dubbed line with Gemini and repairs it, for ~$0.05 per dubbed minute.

How will it solve the problem?

- YouTube: creators who add multi-language audio get over 25% of watch time from other languages. VibeRick Dub makes that track a one-link job instead of a studio booking.

USP

- Own voice + time-fitted translation + a self-verifying agent + consent by design (two locks: channel consent and speaker verification), all on Google Cloud.

List of features offered by the solution

Creator's own voice
Gemini 3.8 Flash TTS voice replication from a 20 s sample + spoken consent

Translation that fits the clock
Each sentence gets a character budget from its time slot and the voice's measured speed

Self-verifying agent
Gemini listens to every dubbed line; too long → shorten, wrong → re-speak → re-translate → flag

Accurate transcript
Chirp 3 timings + Gemini proofreading by listening (7.6% CER, on par with the best tool we tested)

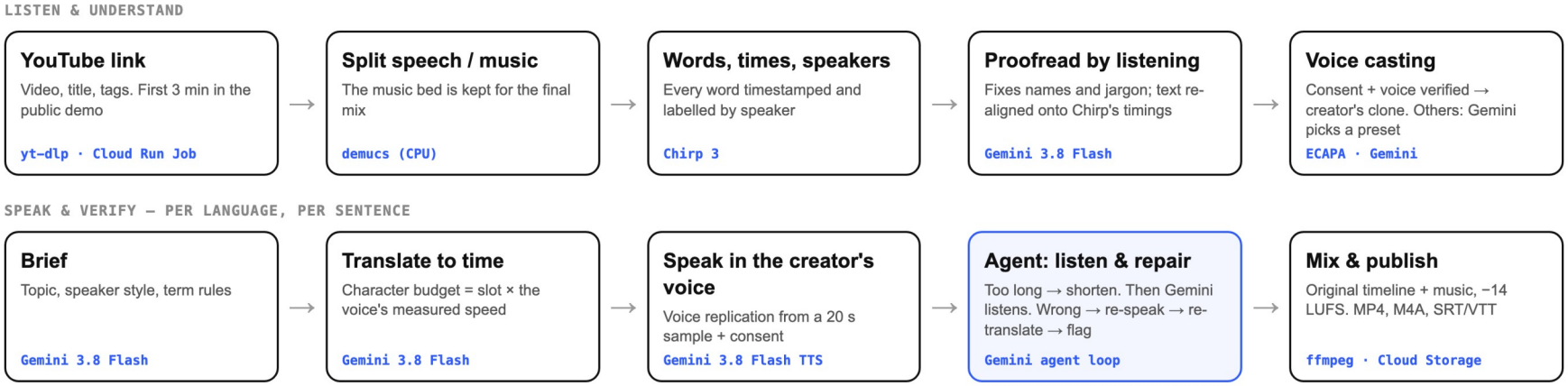
Multi-speaker
Chirp 3 diarization; Gemini casts a fitting preset voice for every other speaker

Consent lock
Clone only for the channel owner whose voice verifies against the consent recording

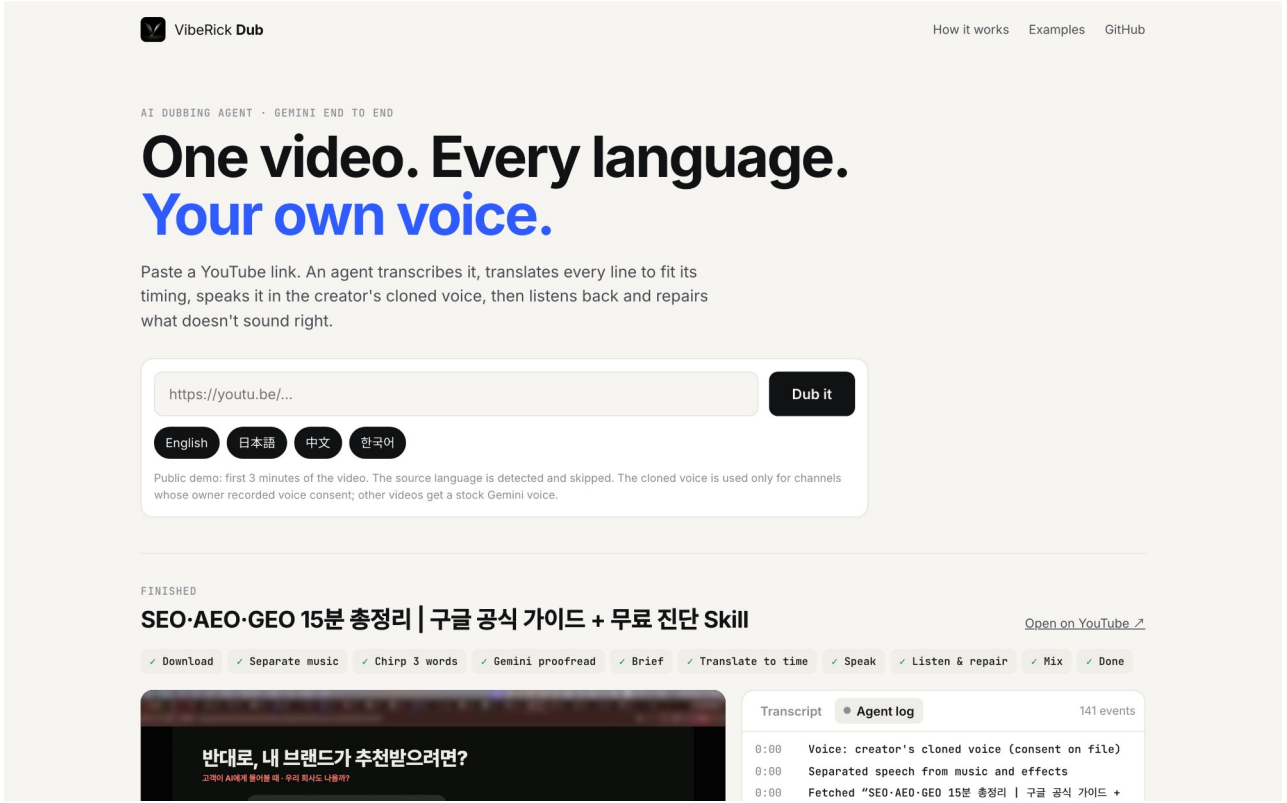
Live studio
Watch the agent work in real time; switch languages on the video; synced transcript

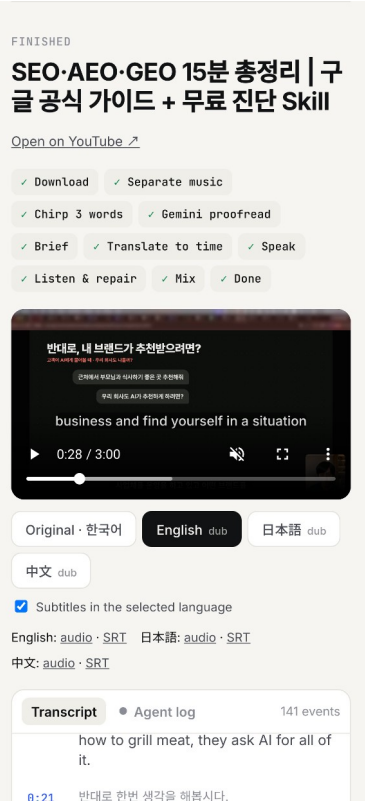
Ready to upload
Dubbed MP4 + M4A (−14 LUFS, original music kept) + SRT/VTT subtitles

Process flow diagram

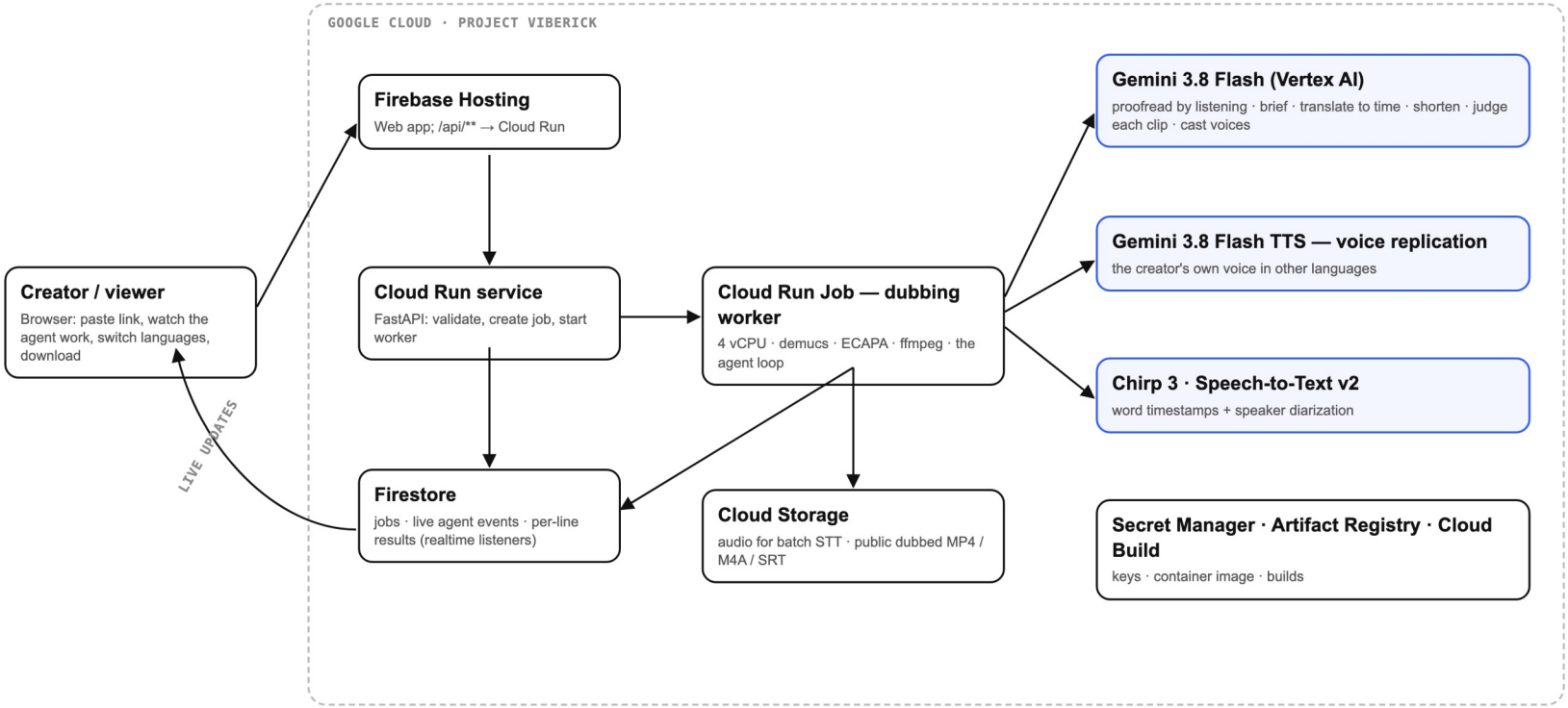


Wireframes / mock: the live product





Architecture diagram of the proposed solution



Technologies used in the solution

Technology	Role
Gemini 3.8 Flash (Vertex AI)	Proofread by listening · dubbing brief · translate to time · shorten · judge every clip · cast voices
Gemini 3.8 Flash TTS — voice replication	The creator's own voice in other languages (SynthID watermarked)
Chirp 3 (Speech-to-Text v2)	Word timestamps + speaker diarization — the timing backbone
Cloud Run (service + Job)	FastAPI API · dubbing worker (4 vCPU, demucs, ECAPA, ffmpeg)
Firestore	Jobs, live agent event stream, per-line results (realtime listeners)
Firebase Hosting	Web app; /api/** rewritten to Cloud Run
Cloud Storage	Batch STT input; public dubbed MP4/M4A/SRT/VTT
Secret Manager · Artifact Registry · Cloud Build	Keys · container image · builds
Open source	demucs (music bed), ECAPA-TDNN (speaker verification for the consent lock), yt-dlp, ffmpeg

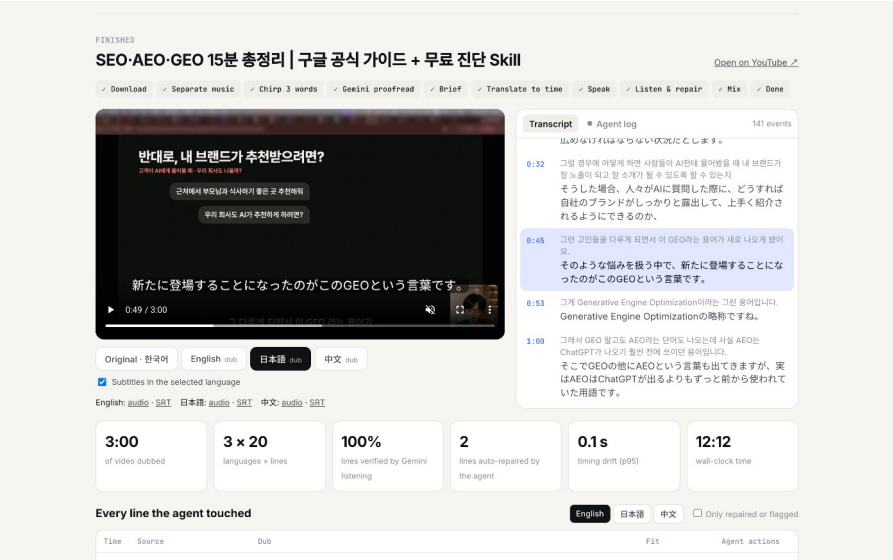
Estimated implementation cost

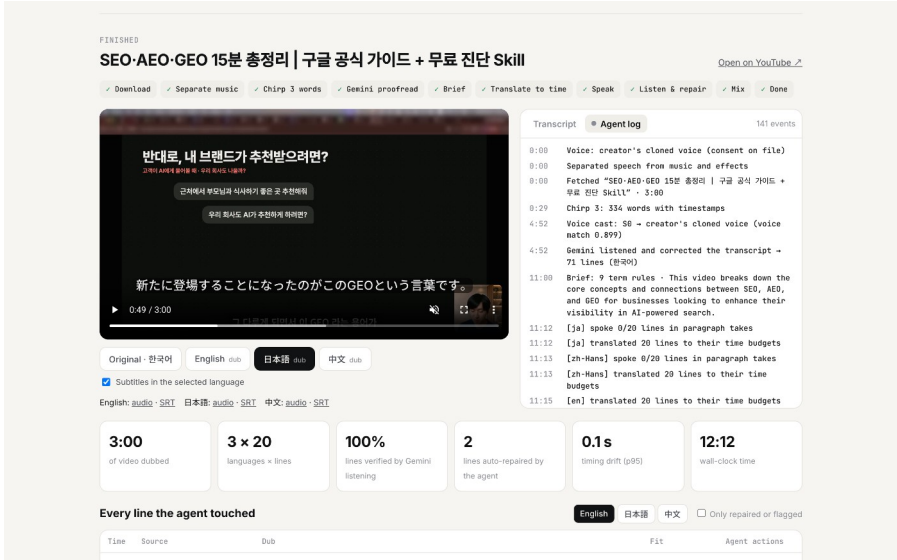
Per dubbed minute (measured)	USD
Gemini 3.8 Flash text + listening checks	0.008
Gemini 3.8 Flash TTS (25 audio tokens/s, $9/1M)	0.014
Chirp 3 (per source min ÷ 3 languages)	0.005
Cloud Run Job (4 vCPU)	0.003
Total	≈ 0.03

Voice-clone dubbing	USD / min
ElevenLabs	0.33 – 2.20
Rask AI	≈ 1.56
Perso AI	≈ 0.55 – 0.73
VibeRick Dub	≈ 0.03

Competitor list prices checked 2026-10-10. Fixed cost: Cloud Run scales to zero; Firestore and Hosting stay within the free tier for the demo.

Snapshots of the prototype





Left: Japanese dub with subtitles and the synced transcript. Right: the agent log — every shorten, re-speak, re-translate and Gemini verdict, streamed live from Firestore.

Prototype performance report / benchmarking

Measure	Result
3-min Korean video → en, ja, zh: lines verified by Gemini listening	69 / 69
Timing drift (p95, worst language)	0.16 s
Wall-clock, 3 languages	2:52
Full 15:05 video → 3 languages: verified · wall-clock	366 / 369 · 10:58
2-speaker English podcast → Korean: lines verified; voices cast	28 / 28 · male + female presets
Transcript CER vs creator's edited captions (3 runs)	Chirp 3 + Gemini 7.6–7.9% · Chirp 8.6% · Gemini 10.6–11.8%
Gemini-only timestamps drift (why Chirp is the clock)	up to 42 s
Speaker similarity to the creator (ECAPA cosine)	English clone 0.68 · creator 0.90 · others 0.00–0.09

Additional details / future development

- More languages: the same path covers every language Gemini TTS speaks; subtitles for 100+.
- One-click publish to YouTube multi-language audio via the YouTube Data API.
- Guest consent in the browser: guests record the consent line and get their own cloned voice.
- Line editor: fix one line, re-speak only that line; glossary per channel.
- Emotion and emphasis transfer from the original delivery; lip-sync for on-camera close-ups.
- Shorts and podcasts: the same agent for vertical video and audio-only feeds.

Links

GitHub public repository

https://github.com/jays0606/viberick-dub

Demo video (under 3 minutes)

(demo video link — added at submission)

Final product

https://viberick-dub.web.app

Consent and safety by design

Two locks before a voice is cloned

1. Channel lock: the channel owner recorded the consent statement.

2. Speaker lock: the diarized speaker's voice verifies against that recording (ECAPA cosine ≥ 0.45).

Everyone else gets a Gemini preset voice chosen by Gemini after listening to them.

Only a 192-number voice embedding is stored, never the audio. Every clip carries SynthID.

Speaker vs consent recording	Cosine
Creator (own video)	0.90
Creator's consent take	0.88
Podcast speaker A	0.09
Podcast speaker B	0.00
Threshold	0.45

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Thank you

